

Designing Reservation Reform Without Mass Unrest

An agent-based simulation of protest mobilization against an income-only quota in India, and of the policy designs that would keep it small

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At a glance

Question. If India replaced caste-based reservation with a single income test (family income below ₹8 lakh), how many people would protest, and which policy designs would keep that number small?

Method. Three pieces fit together. A seat-allocation model, built on the rule India actually uses and on five years of JEE (Advanced) rank lists, estimates who gains and loses IIT seats. A synthetic India of 120,000 agents across 640 districts turns those losses, and the threat to caste recognition, into protest. The protest model is checked against news-event data from four real episodes, including the 2018 Bharat Bandh.

Findings.

- The reform is not a small change. Under the real creamy-layer and EWS rules about 6.8% of Indians lose eligibility, and below-line SC students, who keep eligibility, lose about –72% of their IIT seats to below-line General students.
- How many people would protest cannot be pinned down: the answer moves by more than tenfold with assumptions the data cannot settle. The ranking of policy designs is much more stable than the headcount.
- Keeping caste quotas and adding an income filter inside them changes peak turnout by –97% and is the strongest option in every check.
- Compensating the families who lose eligibility (–8%) and adding seats, as was done for EWS in 2019 (–9%), are the weakest options.
- Among options that keep the income-only end state, a cross-party commission (–58%) and grandfathering (–52%) come next; which is stronger is not settled.
- Internet shutdowns and heavy policing lower the peak somewhat and raise deaths.

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1 The question

India reserves seats in public education and government jobs for Scheduled Castes (SC, 15%), Scheduled Tribes (ST, 7.5%) and Other Backward Classes (OBC, 27%), and since 2019 10% for Economically Weaker Sections (EWS) of the General category. Two of these carry an income test already: OBC families in the “creamy layer” are excluded, and EWS applies only below ₹8 lakh. SC and ST quotas have no income test.

This report asks what would happen if the whole system were converted into a single income test at ₹8 lakh, with caste removed as a criterion:

1. How many people would protest, and for how long?
2. Which changes to the policy’s design, sequencing and politics would keep that protest small?

1.1 Who changes status

An earlier version of this analysis said that only SC and ST families above ₹8 lakh, about 2.9% of Indians, would lose eligibility. That treated the two existing income tests as the same simple cut. They are not. The OBC creamy-layer test ignores salary and farm income, so many OBC families above ₹8 lakh are eligible today and would lose eligibility too. The EWS test also excludes families with land or property above set limits, so some poorer General families are ineligible today and would gain. With central estimates, about 6.8% of Indians lose eligibility and 3.1% gain it.

1.2 Who gains and loses seats

Eligibility is only half the story. Today each category has its own quota. After the reform, all families below the line compete in one pool. The seat-allocation model, which uses the rule India actually applies and the published JEE (Advanced) rank lists for 2021–2025, finds that in IIT admissions:

- below-line SC students would lose –72% of their seats and below-line ST students –85%, although they keep eligibility;
- below-line General students would gain (+71% for those eligible for EWS today, +158% for those who are not), and below-line OBC students would gain a little (+23%).

The reason is that, at the same income, SC and ST candidates sit lower in the rank lists; a pool open to every below-line family therefore admits fewer of them than their own quota does.

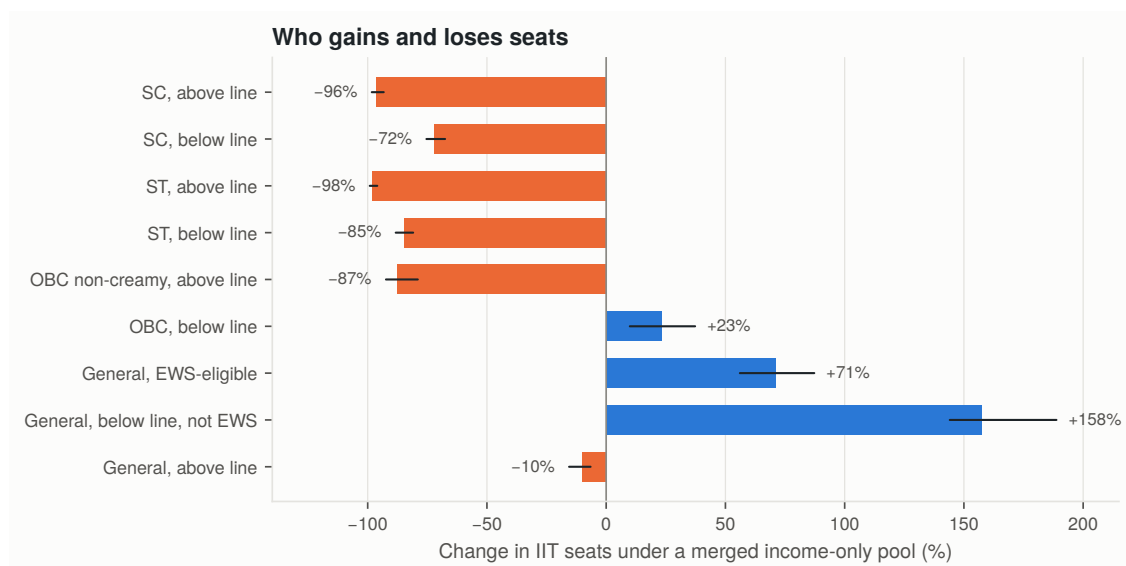


Figure 1: Change in IIT seats by group under a merged income-only pool (median, with 90% intervals over the uncertainty about family incomes).

2 Theory: why people would protest

Five bodies of theory shape the model.

2.1 Loss aversion and the politics of taking things away

People weigh losses roughly twice as heavily as equal gains; Tversky and Kahneman estimate a loss-aversion coefficient of about 2.25. Pierson's work on welfare retrenchment builds on this: cutting an existing benefit is harder than creating one, and governments that succeed use grandfathering, compensation for losers, and blame-sharing through broad coalitions.

In plain words. Taking away a quota feels worse than never having had one, and the people who lose are easier to organize than the people who gain.

2.2 Symbolic threat and group identity

For SC and ST communities, caste-based reservation is also a constitutional recognition of historical injustice. Removing caste as the basis therefore carries a symbolic threat independent of any individual's eligibility. The 2018 bandh, which followed a court ruling that removed no one's quota, is the clearest Indian evidence that symbolic threat alone can mobilize at national scale.

In plain words. Many people would protest not because they lose a seat, but because they read the change as erasing their community's recognition.

2.3 Threshold cascades

Granovetter's threshold model treats each person as having a personal threshold: the share of others who must already be protesting before they join. Small differences in the distribution of thresholds or grievances can produce very large differences in turnout.

In plain words. People join when enough of their neighbours already have. Pushing grievance down a little can remove far more protest than you would expect, because it stops the chain reaction early.

2.4 Organization

Organizations, parties and networks supply the bandh calls and messaging that turn grievance into turnout. But the 2018 bandh was organized with little central leadership or party backing, through local groups and WhatsApp; the 2024 bandh had opposition-party backing and drew a mixed response. Party machinery is one amplifier among several, not a precondition.

In plain words. Anger needs some machinery, but in 2018 the machinery was a phone in every pocket.

2.5 Repression and backfire

Studies of repression find that force sometimes deters and sometimes backfires. Rydzak finds that internet shutdowns in India are associated with more violent collective action, not less.

In plain words. Cutting the internet or sending in heavy policing may shrink the crowd on a given day, and it costs lives. Whether it also makes protest bigger depends on how people react to deaths.

3 The model in simple terms

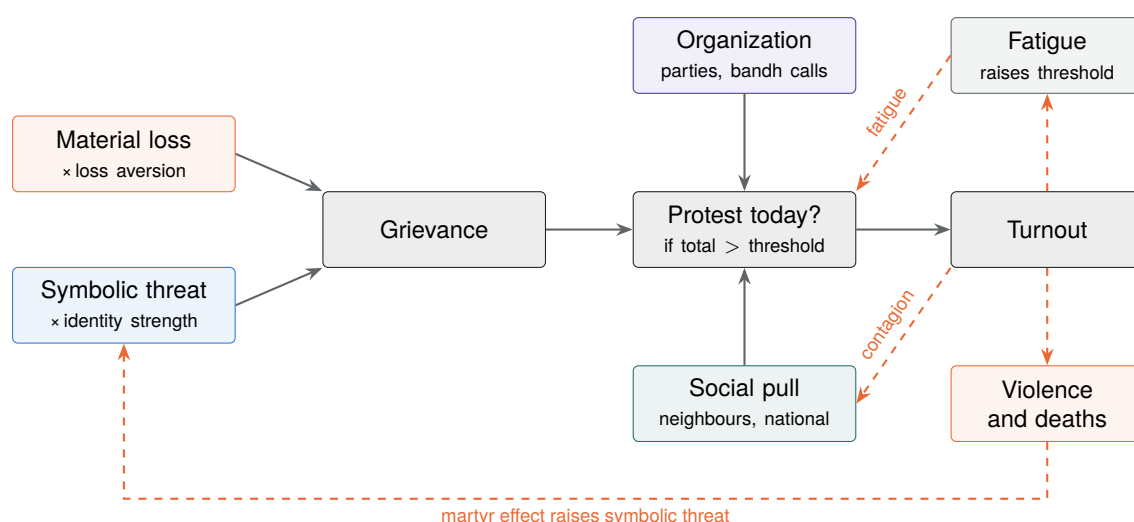


Figure 2: What drives one person's decision each day. Dashed arrows are feedback loops: today's turnout shapes tomorrow's decisions.

3.1 The synthetic India

The model builds 120,000 agents, each standing for about 12,000 people, spread across India's 640 Census 2011 districts in proportion to population. Each district has its own SC and ST shares; OBC and General shares follow the state's figures from the National Family Health Survey. About one family in five sits above ₹8 lakh. Agents live in neighbourhoods of about 100; most neighbourhoods are single-group, and the rest are mixed.

3.2 How one day works

For every agent, every day:

1. **Grievance** is the material loss (weighted 2.25 times if it is a loss), taken from the seat-allocation model, plus the symbolic threat multiplied by how strongly the agent identifies with its group.
2. **Social pull** rises with the share of the agent's neighbourhood that protested yesterday and with its group's national turnout. Both effects saturate.
3. **Organization** adds a push on bandh days. Organizations call a bandh when they expect a credible turnout.
4. The agent **protests** with a probability that rises smoothly as grievance plus pull plus push exceeds its personal threshold. Protesting raises the threshold a little (**fatigue**).
5. Deaths occur in proportion to turnout, most of them on bandh days, and raise symbolic threat for the groups on the street (the **martyr effect**).
6. The government may **concede** under pressure, which relieves grievance but can provoke a counter-mobilization by the General category.

3.3 Checking the model against real protests

Protest turnout in India is rarely counted, and the estimates that exist differ by a factor of a hundred (for 2018, from “thousands” to “hundreds of thousands”). So instead of a headcount, the model is checked against how many protest events the news reported, from the GDELT database, for four episodes: the SC/ST Bharat Bandhs of April 2018 and August 2024, the upper-caste bandh of September 2018, and the EWS amendment of January 2019, which drew little protest. Parameter sets that cannot reproduce all four are discarded; 281 of many thousands survive.

What this shows, honestly:

- The event data tell us how big the episodes were *relative to each other* much better than how big they were. The surviving sets imply that between 7–539 lakh people joined the 2018 bandh.
- When one episode was hidden from the model and predicted from the others, the predictions of deaths and of which states protest were weak. The model is a tool for comparing policy designs, not a forecast.

4 Results at a glance

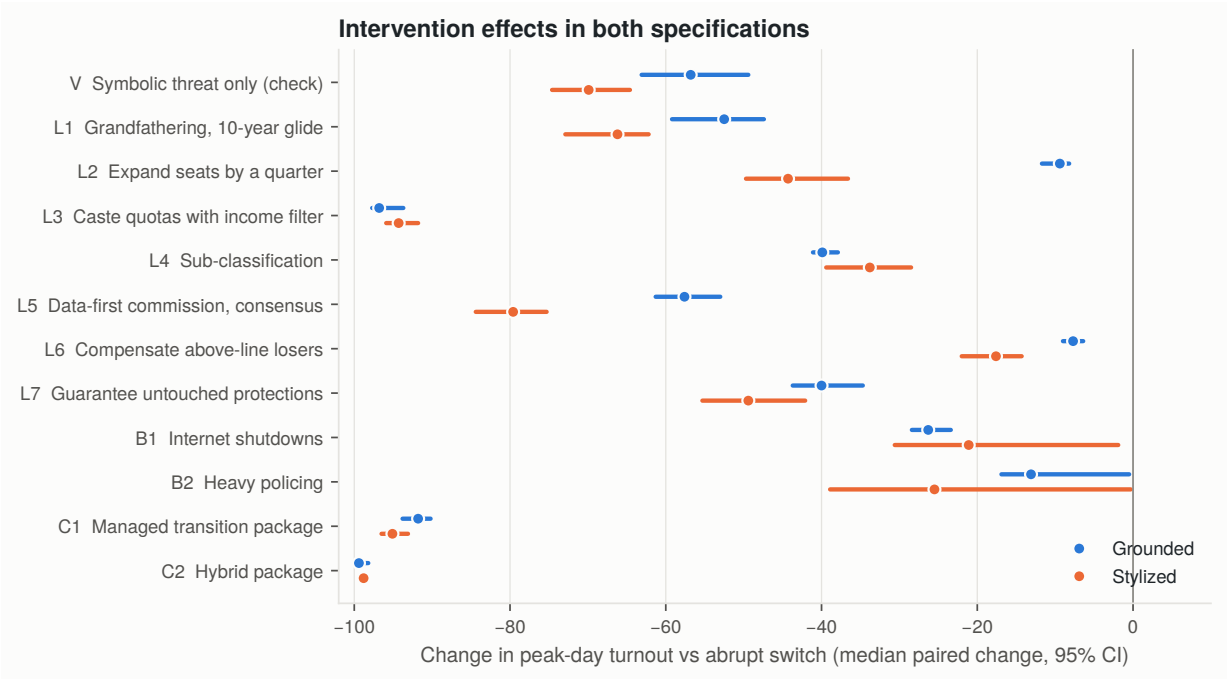


Figure 3: Change in peak-day turnout for each option compared with the abrupt switch in the same simulated world. Dots are medians, lines 95% intervals, for the data-grounded model and the original stylized model.

Table 1: Grounded model, 50 paired runs per scenario. Peak in lakh; change is the median paired change with 95% interval; ever protested in crore.

	Scenario	Peak	10th–90th	Change	95% CI	Ever	Deaths
Base	Abrupt ₹8L income-only switch	214.1	30.6–1248.7	–	–	3.32	95
V	Validation: symbolic threat only (2018-type shock)	64.1	10.6–719.8	–57%	–63 to –49%	1.67	67.5
L1	Grandfather current cohorts + 10-year glide	83.5	13.6–751.2	–52%	–59 to –47%	1.69	70
L2	Expand seats by a quarter (EWS precedent)	193.4	27.1–1193.6	–9%	–12 to –8%	2.99	93.5
L3	Keep caste sub-quotas, add income filter	12.5	1.1–34.1	–97%	–98 to –94%	0.39	10
L4	Sub-classify to favour most-deprived	127.7	25.6–751.4	–40%	–41 to –38%	2.23	99.5
L5	Data-first commission + cross-party consensus	84.4	11.8–700.9	–58%	–61 to –53%	1.57	57.5
L6	Compensate above-line losers	197.0	28.2–1204.4	–8%	–9 to –6%	3.00	95
L7	Guarantee untouched protections	114.0	22.2–846.8	–40%	–44 to –35%	1.92	78.5
B1	Internet shutdowns	146.5	26.5–793.8	–26%	–28 to –23%	2.60	139
B2	Heavy policing and mass arrests	188.5	24.4–1167.5	–13%	–17 to 0%	3.41	219.5
C1	Managed transition package (L1+L2+L5+L6+L7)	18.7	1.9–78.6	–92%	–94 to –90%	0.70	15
C2	Hybrid design package (L3+L4+L1+L5+L6+L7)	1.7	0.4–12.5	–99%	–100 to –98%	0.13	2

Table 2: Stylized model, central regime, 50 paired runs per scenario.

	Scenario	Peak	10th–90th	Change	95% CI	Ever	Deaths
Base	Abrupt ₹8L income-only switch	46.1	14.9–118.9	–	–	1.29	18.5
V	Validation: symbolic threat only (2018-type shock)	13.3	6.0–37.6	–70%	–75 to –65%	0.47	5.5
L1	Grandfather current cohorts + 10-year glide	15.6	6.6–41.5	–66%	–73 to –62%	0.50	5.5
L2	Expand seats by a quarter (EWS precedent)	24.1	9.8–65.8	–44%	–50 to –37%	0.74	11
L3	Keep caste sub-quotas, add income filter	2.9	1.7–4.3	–94%	–96 to –92%	0.10	1
L4	Sub-classify to favour most-deprived	26.9	11.6–71.4	–34%	–39 to –28%	0.83	13.5
L5	Data-first commission + cross-party consensus	8.3	4.1–24.2	–80%	–84 to –75%	0.41	4.5
L6	Compensate above-line losers	34.9	13.1–93.9	–18%	–22 to –14%	1.08	16.5
L7	Guarantee untouched protections	18.2	9.6–55.2	–49%	–55 to –42%	0.68	7.5
B1	Internet shutdowns	42.8	17.2–73.7	–21%	–31 to –2%	1.41	54.5
B2	Heavy policing and mass arrests	32.9	17.4–73.3	–26%	–39 to 0%	1.61	79.5
C1	Managed transition package (L1+L2+L5+L6+L7)	2.2	1.4–3.4	–95%	–96 to –93%	0.12	1
C2	Hybrid design package (L3+L4+L1+L5+L6+L7)	0.6	0.4–0.9	–99%	–99 to –98%	0.04	0

About the headcounts The grounded model's baseline peak of 2.14 crore rests on two things the data cannot settle: how strongly the reform would threaten caste recognition compared with the 2018 ruling, and how much material grievance weighs. Across the plausible range the peak runs from 29.0 lakh to 6.78 crore. Read the headcounts as illustrations and the changes as the result.

5 Where the anger comes from, and who protests

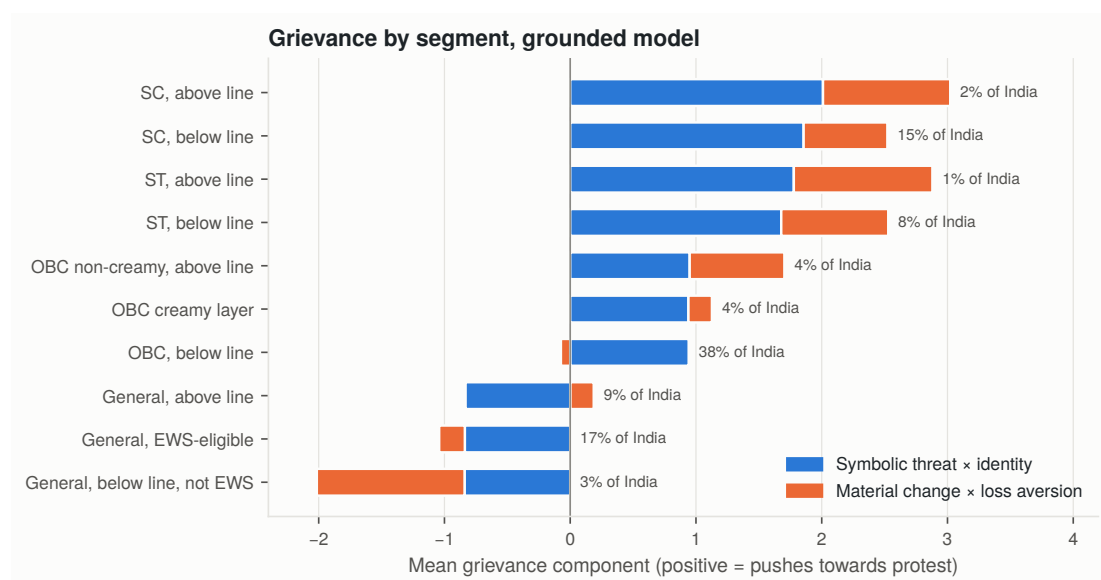


Figure 4: Average grievance by group in the grounded model, split into symbolic threat and material change.

- **Mostly symbolic, but not only.** Symbolic threat is 78% of all the grievance that pushes people towards protest. Removing it cuts the peak by -99% ; removing the material losses cuts it by -57% . For below-line SC and ST families, the seat losses add between a third and a half to their grievance.
- **Who protests.** SC communities make up 63% of protester-days and ST communities 26%.
- **Where.** Uttar Pradesh (15%), Bihar (8%), West Bengal (8%), Madhya Pradesh (7%), Maharashtra (7%). This is where the SC and ST population lives, not a prediction of where organizers will succeed: in 2018, Punjab, where the bandh call began, reported far more protest than its population share.

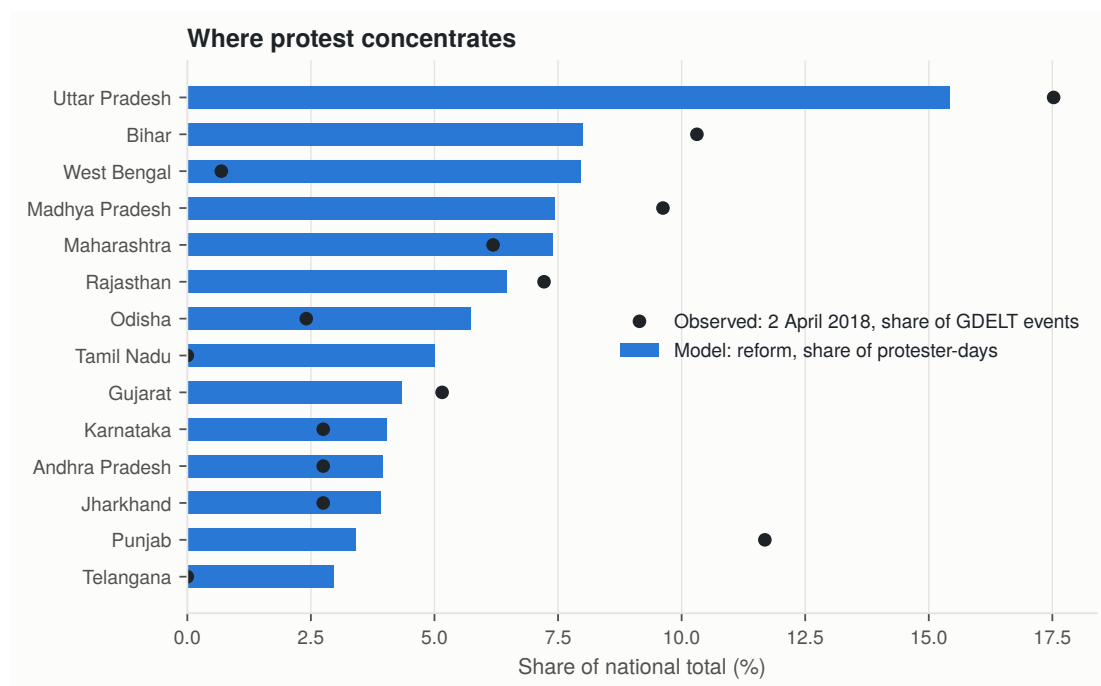


Figure 5: Share of protest by state in the grounded model, and the share of reported protest events by state on 2 April 2018.

5.1 Does the model behave like real protests?

Five patterns from past episodes, and whether the model shows them:

1. A threat to recognition alone can bring out a national bandh: **yes**.
2. Protest concentrates on bandh days: **mostly** (bandh days are about 9 times ordinary days).
3. Party backing is not what makes a bandh large: **yes** (raising party backing from the 2018 to the 2024 level changes the peak by a factor of only 1.04).
4. A group splits when a reform creates winners inside it: **yes** (sub-classification changes the poorest sub-castes' participation by -88%).
5. Some bandhs turn deadly and governments concede: **partly**. Deaths and concessions happen in the model, but it cannot tell which bandhs will turn violent.

6 Scenario by scenario

Each scenario follows the same layout: what the policy is, the theory behind it, a plain-language summary, how it enters the model, the outcome, and why. Changes compare peak-day turnout with the abrupt switch in the same simulated world.

6.1 Base: the abrupt ₹8 lakh income-only switch

What it is

Parliament replaces SC, ST, OBC and EWS quotas with a single income test at ₹8 lakh, effective immediately, with no consultation or transition.

In plain words. This is the reference point: the reform done in the most provocative way.

Outcome

Peak day (grounded)	Change	Ever protested	Deaths
2.14 crore (30.6–1248.7)	–	3.32 crore	95

Stylized specification: 46.1 lakh peak, 1.29 crore ever protest, 18.5 deaths.

Why

SC communities carry most of the turnout: they combine the highest symbolic threat, the strongest organizations and, according to the allocation model, the largest seat losses. In the grounded model the government concedes in 100% of runs, after a median of 5 days.

6.2 V: symbolic threat only (a check)

What it is

All material loss is removed, so only the symbolic threat remains, much like the 2018 ruling.

Outcome

Peak day (grounded)	Change	Ever protested	Deaths
64.1 lakh (10.6–719.8)	–57%	1.67 crore	67.5

Stylized specification: 13.3 lakh (–70%).

Why

Symbolic threat alone still produces a very large protest. Material loss matters more in the grounded model than in the stylized one, because the allocation model shows that below-line SC and ST students lose most of their seats, not a small share.

6.3 L1: grandfather current cohorts, with a 10-year glide

What it is

Students and job applicants already in the pipeline keep their current category rights. The income test is phased in over ten years.

In plain words. Nobody who is counting on a quota today loses it, and the change arrives slowly enough to plan around.

How it is modelled

Every material loss is cut to 30% of its value; symbolic threat falls by 5%.

Outcome

Peak day (grounded)	Change	Ever protested	Deaths
83.5 lakh (13.6–751.2)	–52%	1.69 crore	70

Stylized specification: 15.6 lakh (–66%).

Why

Grandfathering removes most of the material grievance, and in the grounded model material grievance is large. It does nothing about recognition.

6.4 L2: expand seats

What it is

Capacity grows by a quarter, the scale used to introduce EWS in 2019, and the merged pool fills the larger intake.

In plain words. If the pie grows, maybe nobody's slice has to shrink.

How it is modelled

The allocation model is rerun with 25% more seats.

Outcome

Peak day (grounded)	Change	Ever protested	Deaths
1.93 crore (27.1–1193.6)	–9%	2.99 crore	93.5

Stylized specification: 24.1 lakh (–44%).

Why

The EWS precedent does not transfer. EWS added a new category without merging any; here, even with a quarter more seats, the merged pool still ranks SC and ST candidates against General candidates, and below-line SC students still lose –63% of their seats. The stylized model assumed that expansion removes the loss, which is why it looked better there.

6.5 L3: keep caste quotas, add an income filter inside them

What it is

SC, ST and OBC quotas stay, but within each only families below the income line are eligible. In effect this extends the OBC creamy layer to SC and ST. Brazil's 2012 Quota Law combines criteria in a similar way.

In plain words. Keep the caste categories, but stop the richest families inside them from using the quota.

How it is modelled

Symbolic threat for SC, ST and OBC falls to 40% of its value. The allocation model is rerun with SC and ST quotas closed to above-line families: their seats go to below-line SC and ST students, who gain (+32% for SC, +37% for ST).

Outcome

Peak day (grounded)	Change	Ever protested	Deaths
12.5 lakh (1.1–34.1)	–97%	0.39 crore	10

Stylized specification: 2.9 lakh (–94%).

Why

It removes most of both drivers at once: the symbolic threat, and the seat losses of the below-line majority, which become gains. It is the strongest option in every check. Two cautions: the 2024 bandh, triggered by a mere suggestion of an SC/ST creamy layer, suggests that the symbolic relief may be smaller than assumed; and this design keeps caste as a criterion, so it changes the original question.

6.6 L4: sub-classify to favour the most-deprived

What it is

Within SC and ST, a larger share of benefits goes to the most-deprived sub-castes, as permitted by the Supreme Court in 2024.

In plain words. Give the poorest sub-castes a better deal, and many of them will not join the protest.

How it is modelled

The most-deprived tier's material loss falls by half a unit, and its symbolic threat falls from 0.8 to 0.3 times its group's.

Outcome

Peak day (grounded)	Change	Ever protested	Deaths
1.28 crore (25.6–751.4)	–40%	2.23 crore	99.5

Stylized specification: 26.9 lakh (–34%).

Why

The most-deprived tier largely drops out, but the better-off tier keeps protesting.

6.7 L5: data-first commission with cross-party consensus

What it is

The reform goes through a statutory commission that works from the caste census data before any legislation, with opposition parties represented.

In plain words. If every major party signs off, and the design is built on data everyone has seen, the reform reads less as an attack.

How it is modelled

The opposition-party amplifier falls from 1.5 to 1.0; symbolic threat falls by 20%.

Outcome

Peak day (grounded)	Change	Ever protested	Deaths
84.4 lakh (11.8–700.9)	–58%	1.57 crore	57.5

Stylized specification: 8.3 lakh (–80%).

Why

Most of the effect comes from the assumed cut in symbolic threat, not from withdrawing party machinery; in 2018 protest was large without it. Whether this option beats grandfathering depends on how much a commission would really change what the reform means to people, which the model cannot tell.

6.8 L6: compensate the families who lose eligibility

What it is

Families that lose eligibility receive scholarships, hostel places and transitional grants.

In plain words. Pay the people who lose the quota.

How it is modelled

The losses of above-line SC, ST and non-creamy OBC families are halved.

Outcome

Peak day (grounded)	Change	Ever protested	Deaths
1.97 crore (28.2–1204.4)	–8%	3.00 crore	95

Stylized specification: 34.9 lakh (–18%).

Why

It reaches a small minority, not the below-line majority whose seats the merged pool takes, and money does nothing about symbolic threat.

6.9 L7: guarantee untouched protections

What it is

The reform is written with explicit guarantees about what stays untouched: SC/ST reserved seats in legislatures, the Atrocities Act, and a scheduled review clause.

In plain words. Say clearly, in law, what is not changing.

How it is modelled

Symbolic threat falls by 15%.

Outcome

Peak day (grounded)	Change	Ever protested	Deaths
1.14 crore (22.2–846.8)	–40%	1.92 crore	78.5

Stylized specification: 18.2 lakh (–49%).

Why

A 15% cut in symbolic threat produces a much larger cut in turnout, because the cascade multiplies any reduction.

6.10 B1: internet shutdowns (tested for backfire)

What it is

Mobile internet is suspended in areas with protests.

How it is modelled

Neighbourhood and national influence fall by 40% and bandh mobilization by 20%, while violent deaths rise 2.5 times.

Outcome

Peak day (grounded)	Change	Ever protested	Deaths
1.47 crore (26.5–793.8)	–26%	2.60 crore (–20%)	139 (×1.4)

Stylized specification: 42.8 lakh (–21%), deaths ×2.5.

Why

Weaker coordination trims the peak. The rise in deaths is mostly the assumption itself: shutdowns are modelled as making protest more violent, following Rydzak's evidence.

6.11 B2: heavy policing and mass arrests (tested for backfire)

What it is

Large deployments, preventive detentions and forceful dispersal on bandh days.

How it is modelled

Turning out on bandh days costs more; deaths attributed to the police rise so that total deaths triple; each death raises symbolic threat half as much again.

Outcome

Peak day (grounded)	Change	Ever protested	Deaths
1.89 crore (24.4–1167.5)	–13%	3.41 crore (–4%)	219.5 (×2.6)

Stylized specification: 32.9 lakh (–26%), +14% ever protest, deaths ×3.9.

Why

Policing lowers the peak a little and costs many lives. In the stylized model it also makes more people protest in total, because martyrs keep protest going between bandhs. In the grounded model it does not, because the government concedes within days. Whether force backfires on turnout depends on how people react to deaths, which none of the episodes can tell us; that it costs lives does not.

6.12 C1: the managed-transition package

What it is

Keep the income-only end state, but get there with seat expansion (L2), grandfathering (L1), a consensus commission (L5), compensation (L6) and guarantees (L7).

Outcome

Peak day (grounded)	Change	Ever protested	Deaths
18.7 lakh (1.9–78.6)	–92%	0.70 crore	15

Stylized specification: 2.2 lakh (–95%).

Why

Each component pushes people further below their thresholds, and because turnout responds non-linearly the combination does far better than its parts.

6.13 C2: the hybrid-design package

What it is

L3 with sub-classification (L4), grandfathering (L1), the commission (L5), compensation (L6) and guarantees (L7).

Outcome

Peak day (grounded)	Change	Ever protested	Deaths
1.7 lakh (0.4–12.5)	–99%	0.13 crore	2

Stylized specification: 0.6 lakh (–99%).

7 How sure are these results?

- **How strongly each policy works is uncertain.** When every policy's strength is drawn at random from a wide range, keeping caste quotas with an income filter still comes first with probability 95%. The commission beats grandfathering with probability 34%, so grandfathering is more often the stronger of the two.
- **The model's structure is uncertain.** Across 11 alternative structures (different threshold shapes, organizations that call bandhs only when turnout is credible, a government that can concede, mixed neighbourhoods, deaths that deter instead of mobilize), L3 comes first in 11 and compensation last in 11.
- **Every parameter moved at once.** A Morris screen across fifteen parameters finds L3 ahead of the commission in 98% of design points in the stylized model and 100% in the grounded one.
- **What the data cannot settle.** Across the reform's shock size and party backing, the baseline peak moves from 29.0 lakh to 6.78 crore, but L3 comes first in all 6 combinations.

8 What this means

1. **A merged income pool is not neutral.** It moves most seats of below-line SC and ST students to below-line General students, and adding seats on the EWS model does not undo that.
2. **Design matters more than compensation.** A design that keeps caste as the qualifier while filtering by income does far better than paying the people who lose.
3. **If the income-only end state is the goal,** phasing and a process that changes what the reform means matter most, and they compound.
4. **Suppression costs lives.** It lowers the peak somewhat; whether it also broadens protest is uncertain.

8.1 Limitations

- How strongly a commission, a guarantee or an income filter reduces the sense of threat is a judgement, handled with wide ranges.
- News-event data pin down the relative size of past protests, not their headcounts. The reform's protest level is therefore uncertain by more than tenfold.
- The seat model covers IIT admissions only. Government jobs and other institutions may differ.
- Jati-level differences, state politics, courts, elections and media are not modelled.
- Out-of-sample predictions of deaths and of which states protest were weak.

9 Reproducing this report

```

pip install -r requirements.txt
python -m experiments.merged_pool_allocation --draws 300
python -m experiments.episode_calibration --wave-size 6000 --waves 3
python -m experiments.compare_interventions --specification grounded
python -m experiments.compare_interventions --regime central
python -m visualization.render_grounding_figures
python -m visualization.write_latex_result_macros
python -m visualization.write_grounding_macros
cd report && xelatex research_report.tex && xelatex research_report.tex

```

The paper's Reproducibility section lists every command.

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